

JSON with JSONPath

JSONPath is a query language for JSON with features similar to XPath for XML. JSONPath is used for selecting and extracting a sub-section from the JSON document.



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JSONPath is a query language for **JSON** with features similar to **XPath** for XML. JSONPath is used for selecting and extracting a sub-section from the JSON document.

1. JSONPath Dependencies

To use JSONPath, we will need to include its dependency and then use it.

Language	Dependency
JavaScript	JSONPath JavaScript File
Node	<code>npm install JSONPath</code>
PHP	JSONPath PHP include file
Python	<code>pip install jsonpath-rw</code>
Java	json-path

2. JSONPath Syntax

A JsonPath expression begins with the dollar sign (\$) character, which refers to the root element of a query. The dollar sign is followed by a sequence of child elements, which are separated via dot (code) notation or via the square brackets (code).

The important *JSONPath syntax rules* are:

- \$ symbol refers to the **root object** or element.
- @ symbol refers to the **current object** or element.
- . operator is the dot-child operator, which you use to denote a **child element** of the current element.
- [] is the subscript operator, which you use to denote a **child element of the current element (by name or index)**.
- * operator is a wildcard, returning **all objects** or elements regardless of their names.
- , operator is the union operator, which returns the **union of the children** or indexes indicated.
- : operator is the **array slice operator**, so you can slice collections using the syntax `[start:end:step]` to return a subcollection of a collection.
- () operator lets you pass a **script expression** in the underlying implementation's script language. It's not supported by every implementation of JSONPath, however.
- ? () to *query all items that meet a certain criteria*.

3. JSONPath Expression Examples

Below given are few examples of JSONPath.

```
$.store.book[0].title  
$.store.book[*].title  
$..book[3]
```

```
//or using brackets
```

```
$['store']['book'][0]['title']  
$['store']['book']['*']['title']  
$..['book'][3]
```

```
$.store.book[?(@.price < 10)].title
```

4. JSONPath Example using JavaScript

We have the following JSON document. We will apply the JSONPath expressions to it.

```
{  
  "store":  
  {  
    "book": [  
      {  
        "category": "reference",  
        "author": "Nigel Rees",  
        "title": "Sayings of the Century",  
        "price": 8.95  
      },  
      {  
        "category": "fiction",  
        "author": "Evelyn Waugh",  
        "title": "Sword of Honour",  
        "price": 12.99  
      },  
      {  
        "category": "fiction",  
        "author": "Herman Melville",  
        "title": "Moby Dick",  
        "isbn": "0-553-21311-3",  
        "price": 8.99  
      },  
      {  
        "category": "fiction",  
        "author": "J. R. R. Tolkien",  
        "title": "The Lord of the Rings",  
        "isbn": "0-395-19395-8",  
        "price": 22.99  
      }  
    ]  
  }  
}
```

```

    }
    ],
    "bicycle": {
        "color": "red",
        "price": 19.95
    }
}
}

```

Example 1

Using JSONPath to find the names of all authors.

```
var response = jsonPath(store , "$..author").toJSONString();
```

Program output:

```

[
  "Nigel Rees",
  "Evelyn Waugh",
  "Herman Melville",
  "J. R. R. Tolkien"
]

```

Example 2

Using JSONPath to find the details for book number 4. The array index is zero-based.

```
var response = jsonPath(store , "$..book[3]").toJSONString();
```

Program output:

```

[
  {
    "category": "fiction",

```

```
[{"author": "J. R. R. Tolkien",  
  "title": "The Lord of the Rings",  
  "isbn": "0-395-19395-8",  
  "price": 22.99  
}]
```

Comments

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21 COMMENTS



Most Voted ▼

Rahul

🕒 6 years ago

Can i transform the json string into another format. I want the JSON string to be transform into array of array where property name in book becomes the column names. ?

+ 10 - ➡ Reply

Santosh

🗨 Reply to [Rahul](#) 🕒 1 year ago

Convert to map as json payload stores the values in key and value pair. That is the best approach.

+ 0 - ➡ Reply

Abhineet

🕒 6 years ago

```
public static List readJsonFileDynamic(String filePath, String jsonPath) {  
  
    System.out.println("jsonpath - "+jsonPath);  
    try  
    {  
        String content = new String(Files.readAllBytes(Paths.get(filePath)));  
        Configuration conf = Configuration.builder()  
            .jsonProvider(new GsonJsonProvider())  
            .mappingProvider(new GsonMappingProvider())  
            .build();  
  
        DocumentContext context = JsonPath.using(conf).parse(content);  
        categories = context.read(jsonPath, List.class);//List  
  
    } catch (IOException e) {  
        // TODO Auto-generated catch block  
        e.printStackTrace();  
    }  
    return categories;  
}
```

+ 3 — Reply

Noha

🕒 6 years ago

Can i use jsonpath in url encoding?

+ 3 — Reply

poornima

🕒 3 years ago

how can i get the string value from the array

input ::

```
{“characteristics”: [  
{  
  “name”: “ICCID”,  
  “valueType”: “string”,  
  “value”: “8925102410460008907”  
},
```

```
{
  "name": "IMSI",
  "valueType": "string",
  "value": "636024100008907"
}
]
```

output ::

```
"8925102410460008907"
```

but its coming like when i tried with \$.characteristics[?(@.name=="ICCID")].value

```
[
  "8925102410460008907"
]
```

+ 0 — ➤ Reply

Mohit

🕒 3 years ago

Can I match a top level attribute value

```
{
  "category": "reference",
  "author": "Nigel Rees",
  "title": "Sayings of the Century",
  "price": 8.95
}
```

Need to verify, "category": "reference", is present in json or not.

I tried \$.[?(category == 'reference')]

+ 0 — ➤ Reply

VISHAL

🕒 3 years ago

Can we provide multiple filters like

Book [(category = fiction and

Author = Evelyn Waugh) and (category= reference and author= Nigel Rees)]

+ 0 — ➤ Reply

Lokesh Gupta

🗨 Reply to [VISHAL](#) 🕒 3 years ago

Yes, it is possible. I quickly tried it on this [tool](#).

```
$.book[?((@.category=='fiction' && @.author=='Evelyn Waugh') ||
(@.category=='reference' && @.author=='Nigel Rees'))]
```

+ 0 — ➔ Reply

kiruba

🕒 6 years ago

This looks like the original passport JSON data. Please remove it as it violates the person's privacy.

+ 0 — ➔ Reply

Teja

🕒 6 years ago

```
[
  {
    "firstName": "F1",
    "lastName": "L1",
    "address": [
      {
        "city": "Atlanta",
        "country": "US"
      },
      {
        "city": "Maidson",
        "country": "US"
      }
    ]
  },
  {
    "firstName": "F2",
    "lastName": "L2",
    "address": [
      {
        "city": "Nashville",
        "country": "US"
      },
      {
        "city": "Vegas",
        "country": "US"
      }
    ]
  },
  {
```



```
"firstName": "F3",
"lastName": "L3",
"address": [
  {
    "city": "Minneapolis",
    "country": "US"
  },
  {
    "city": "Nashville",
    "country": "US"
  }
]
}
```

From the above data, how can I get person details who lived in Nashville for example. I am expecting the following result

```
[
  {
    "firstName": "F2",
    "lastName": "L2",
    "address": [
      {
        "city": "Nashville",
        "country": "US"
      },
      {
        "city": "Vegas",
        "country": "US"
      }
    ]
  },
  {
    "firstName": "F3",
    "lastName": "L3",
    "address": [
      {
        "city": "Minneapolis",
        "country": "US"
      },
      {
        "city": "Nashville",
```

```
        "country": "US"
    }
]
}
```

Thank you!

+ 0 — ➡ Reply

ksk

🗨 Reply to [Teja](#) ⌚ 6 years ago

```
$.[?(@.city=="Nashville")]^^^^
```

Looks like this is working fine.

+ 4 — ➡ Reply

Sai

⌚ 6 years ago

Can i write regular expression to match the string in javascript. i saw it working in jay way which is for java but in javascript its not working.

+ 0 — ➡ Reply

Lokesh Gupta

🗨 Reply to [Sai](#) ⌚ 6 years ago

Yes, `string.match(regex)` takes regular expression in javascript.

+ 1 — ➡ Reply

nshah

⌚ 6 years ago

How do I write back to json file using JSONPath ?

+ 0 — ➡ Reply

John

🗨 Reply to [nshah](#) ⌚ 6 years ago

nshah I don't think it is possible to write json elements using JSONPath. Otherwise, it would surely be listed on this page.

+ 0 — ➡ Reply

Test

🕒 6 years ago

```
{
  "kind": "storage#policy",
  "resourceId": "test1",
  "bindings": [
    {
      "role": "roles/admin",
      "members": ["allUsers"]
    },
  ]
}
```

How can I check if members contain 'allUsers'

+ 0 — ➡ Reply

Lokesh Gupta

🗨 Reply to [Test](#) 🕒 6 years ago

Create XPath expression with help of information given in this [xpath example](#).

+ 0 — ➡ Reply

Aparjeet

🗨 Reply to [Lokesh Gupta](#) 🕒 1 year ago

```
[
  {
    "_id": "6537fc3015f82f921221f0fb",
    "bagNumber": "1a",
    "bagType": "cash",
    "amount":
    {
      "$numberDecimal": "4809.52"
    }
  }
]
```

Can i get help to extract the value from above json

I am expecting the output as

```
"_id": "6537fc3015f82f921221f0fb"
```

+ 0 — ➡ Reply

Kanchan

Reply to [Test](#) 5 years ago

`$.[?(@.members=="allUsers"]`

+ 0 - Reply

sarika

7 years ago

How can i append two string response from json using jsonpath ?

Ex. JsonResponse

```
{
  "passport": {
    "page_1": {
      "passport_number": "xxxxxxx",
      "photo": &quot;comment image",
      "sign": &quot;comment image",
      "surname": "RAMADUGULA",
      "name": "SITA MAHA LAKSHMI",
      "nationality": "INDIAN",
      "sex": "",
      "date_of_birth": "23/09/xxxx",
      "place_of_birth": "GUNDUGOLANU",
      "date_of_issue": "11/10/xxxx",
      "date_of_expiry": "10/10/xxxx"
    }
  }
}
```

I just want to append name and surname like **SITA MAHA LAKSHMI RAMADUGULA**.

When i tried using `$['passport']['page_1']['name','surname']` its giving me value :

```
{
  "name" : "SITA MAHA LAKSHMI",
  "surname" : "RAMADUGULA"
}
```

that i don't want. Any idea about it ??

+ 0 - Reply

Debi

Reply to [srika](#) 7 years ago

```
$.passport.page_1.(name & ' ' & surname)
```

Result : "SITA MAHA LAKSHMI RAMADUGULA"

+ -2 - Reply

PREVIOUS

JSON with Ajax

NEXT

JSON Schema

```
"string-key": "john",
```

DO NOT SELL OR SHARE MY PERSONAL INFORMATION